

Name: _____

Date: _____

Complete this quiz before the next workshop. You may refer to the Python Primer handout — the goal is to confirm you've read and understood it, not to test memorization.

Section 1: Basic Recall

Questions 1 – 8

1. What is always the first import in a PyKit program?

2. What is the pattern for importing a class from a module?

3. What is it called when you write `px = NeoPixels()` to set up a hardware object?

4. What does `while True: do?`

5. How many spaces does PyKit convention use for indentation?

6. What does `time.sleep(0.1)` do, and how long does it pause?

7. What module and class would you use to read tilt and motion?

8. What module and class would you use to control RGB LEDs?

Section 2: Understanding

Questions 9 – 14

9. Write the line of code that sets all NeoPixels to red.

10. What is wrong with this code?

```
while True:
    px.fill(Colors.RED)
px = NeoPixels()
```

11. What does the f in an f-string do? Give an example using a variable called temperature.

12. What error would you get if you wrote `From neopixels import NeoPixels`?

13. If you see `NameError: name 'imu' is not defined`, what most likely went wrong?

14. You want to sense proximity gestures and play a sound in response. Which two modules do you combine?

Section 3: Application

Questions 15 – 18

These questions require you to write or reason about code you haven't seen before.

15. Write a complete program that turns all NeoPixels green when the board is tilted UP, and turns them off otherwise. Include all four patterns from the Primer.

16. What is the core problem with using `time.sleep()` when you want two things to happen at the same time? How does `AsyncRunner` solve it?

17. What are the two changes you must make to a regular function to make it work with `AsyncRunner`?

18. (Stretch) Explain in your own words what `await` means in the context of `AsyncRunner`. Use the 'worker' analogy from the Primer if it helps.
